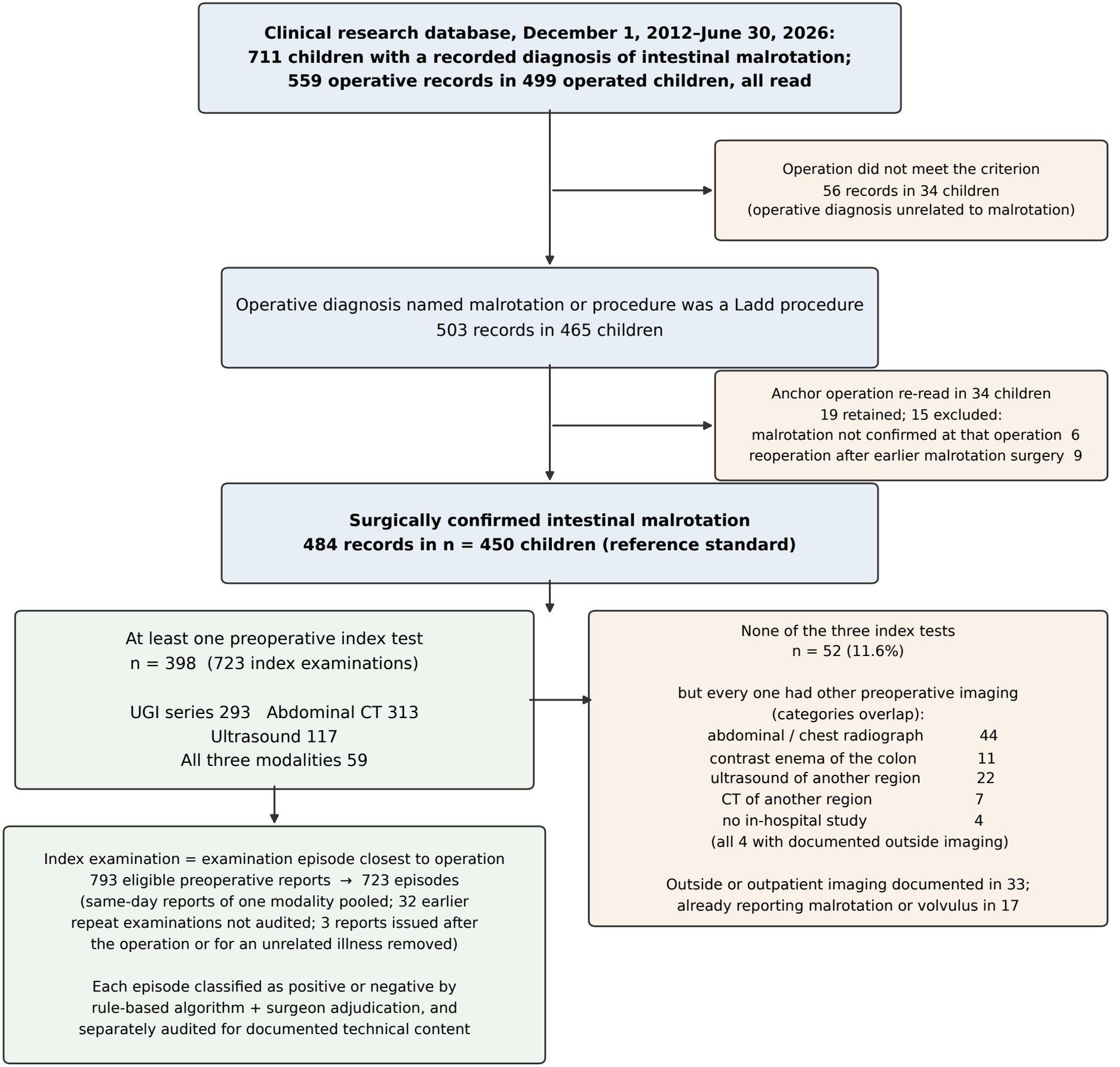
Clinical research database, December 1, 2012–June 30, 2026:
711 children with a recorded diagnosis of intestinal malrotation;
559 operative records in 499 operated children, all read
Operation did not meet the criterion
56 records in 34 children
(operative diagnosis unrelated to malrotation)
Operative diagnosis named malrotation or procedure was a Ladd procedure
503 records in 465 children
Anchor operation re-read in 34 children
19 retained; 15 excluded:
malrotation not confirmed at that operation 6
reoperation after earlier malrotation surgery 9
Surgically confirmed intestinal malrotation
484 records in n = 450 children (reference standard)
At least one preoperative index test
n = 398 (723 index examinations)
UGI series 293 Abdominal CT 313
Ultrasound 117
All three modalities 59
None of the three index tests
n = 52 (11.6%)
but every one had other preoperative imaging
(categories overlap):
abdominal / chest radiograph 44
contrast enema of the colon 11
ultrasound of another region 22
CT of another region 7
no in-hospital study 4
(all 4 with documented outside imaging)
Outside or outpatient imaging documented in 33;
already reporting malrotation or volvulus in 17
Index examination = examination episode closest to operation
793 eligible preoperative reports → 723 episodes
(same-day reports of one modality pooled; 32 earlier
repeat examinations not audited; 3 reports issued after
the operation or for an unrelated illness removed)
Each episode classified as positive or negative by
rule-based algorithm + surgeon adjudication, and
separately audited for documented technical content